Engineering Portfolio

Neev Zeroni

About Me

- Mechanical engineering at University of Minnesota
- 4 years in med tech internships
  - Abbott
  - Veranex
  - 2 different startups
- Skiing/biking
- D&D, MTG
- Video Games







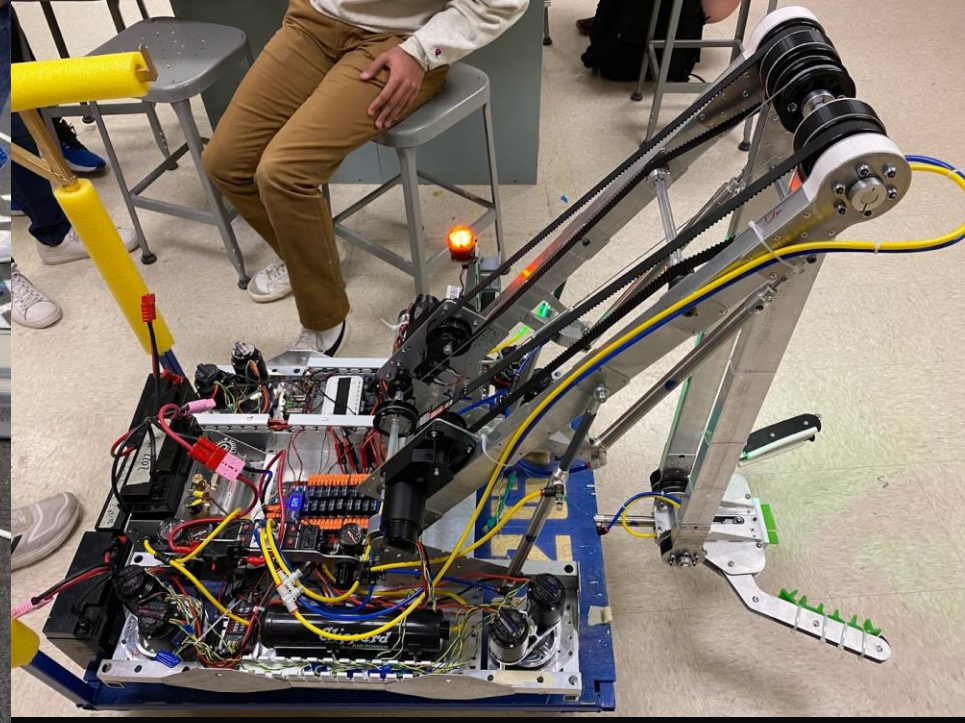


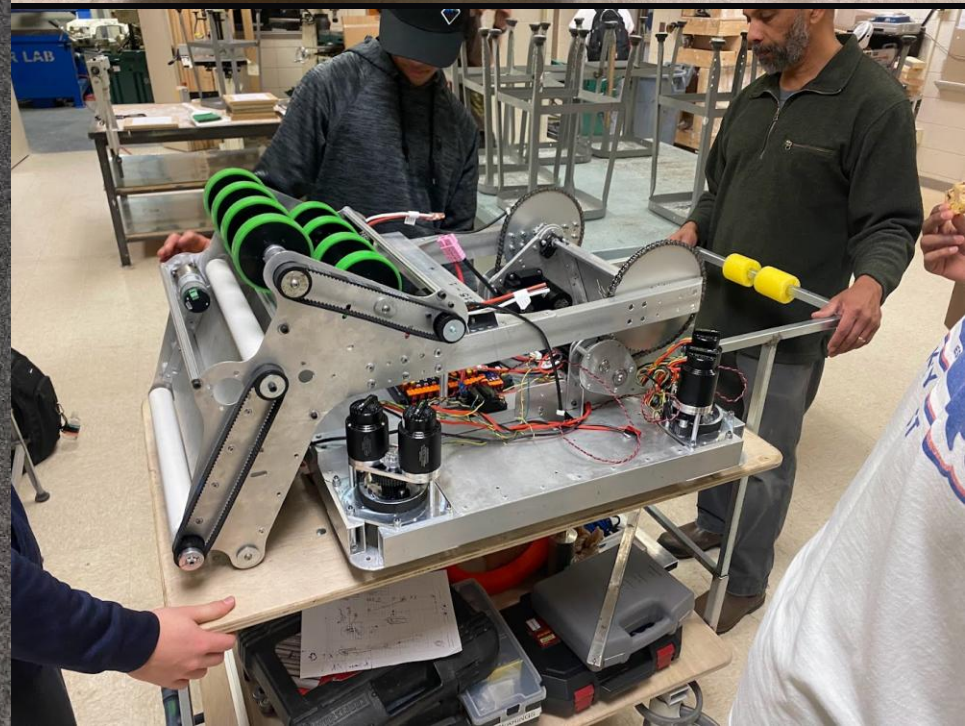
FRC – First Robotics

Mechanical Captain

- Lead the design and manufacturing of 3 different robots
  - T-shirt Cannon
    - Created a tanks drive wheelbase
    - Used 2 pumps to fill 3 tanks, used solenoids to launch shirts stuffed in launch tubes
  - Crescendo Robot
    - Large cantilever arm with intake and flywheels
    - First time implementing swerve drive
  - Charged Up Robot
    - Managed pneumatic air system that controlled large arm and gripper actuation







Northstar Robotics – UMN Robotics Club

Turret lead

- Lead the design, machining, and assembly of the top section of three separate robots for the arc robotics challenge:
  - Standard (teleoperated), Sentry (autonomous), and Hero (Teleop, larger projectiles)
- Designed and tested multiple turbine types and optimized geometry to prevent jamming
- Lead hardware integration in physical design, optimizing space use to fit large amounts of electronics into small spaces



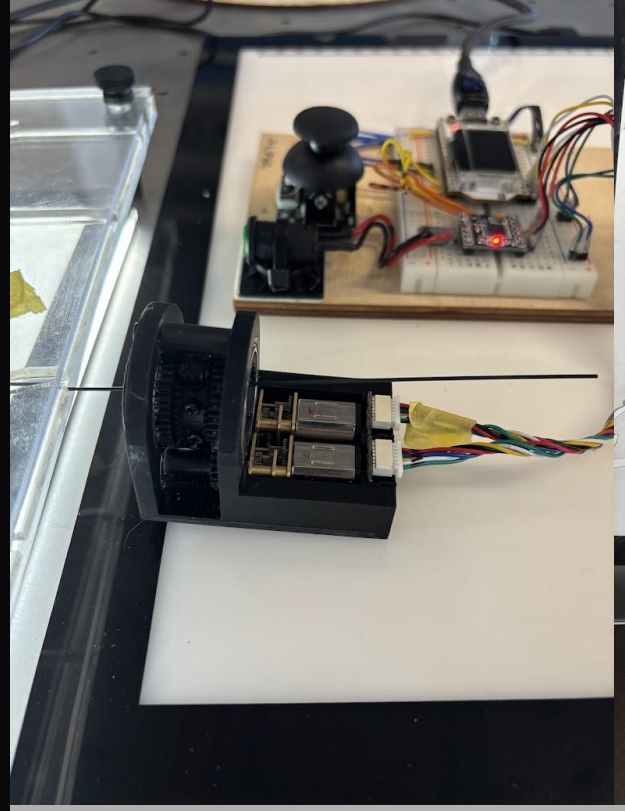




UMN Medical Robotics Lab

Research Assistant

- Designed and created an automatic guidewiring robot running on a 2dof differential system
- Currently working on AI integration and future improvement (4dof)
- Integrating EM tracking localization to be able to autonomously track through a 3d track



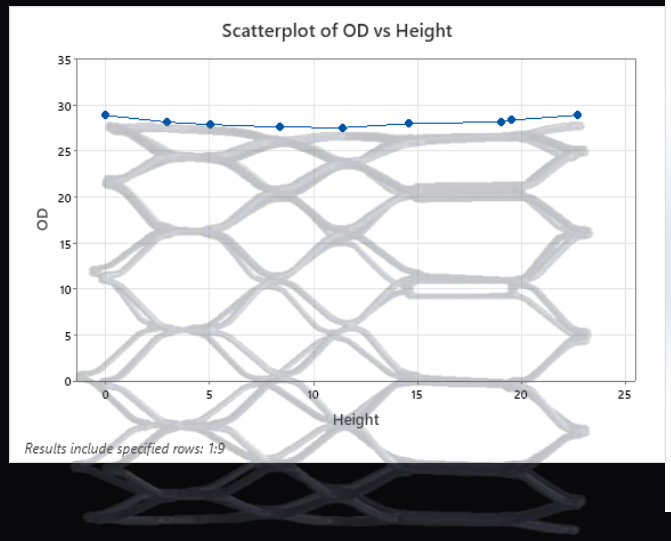


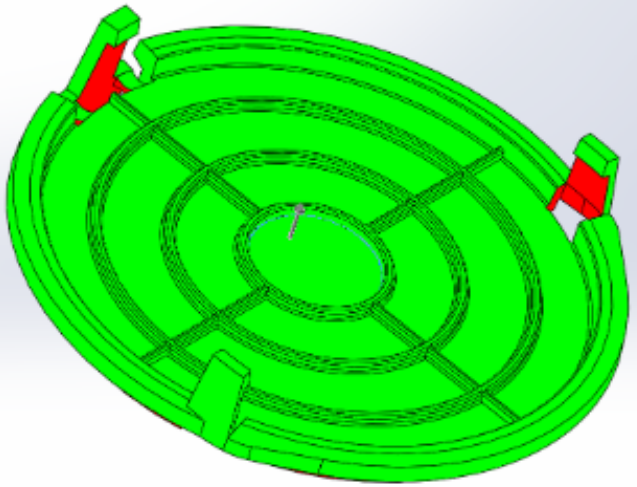


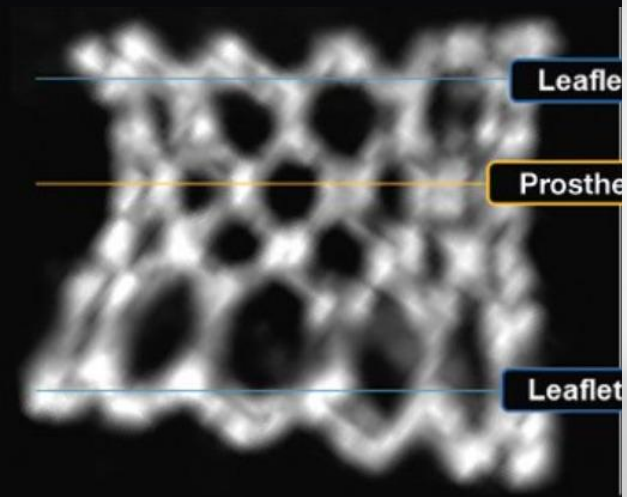
Professional Experience

4 years of mechanical medical device internships

- Co-op at Abbott
  - Worked on a TAVR heart valve prosthesis, lead design studies on stent characterization
  - Optimized design for uniform expansion
  - Lead cadaver study for in-vitro to in-vivo offset
- Veranex
  - Created fixtures for EV1 assembly line builds
  - Streamlined prototype creation process for other engineers











Personal Projects

- 5.3L LS V8 rebuild
- Futura inverted pendulum balancer
- Taco Bell compass
- Generative quadruped design
- Robotic hand

